# Asad Fareed

Lahore, Punjab Pakistan
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Senior Mobile Application Developer with 6+ years of experience building enterprise-grade Android and Flutter applications. Expertise in Automatic Fare Collection (AFC), Open-Loop and Closed-Loop Payment Systems, NFC Technologies, EMV Contactless Payments, MPGS Integration, Validator Software, Facial Recognition, and Anti-Spoofing Solutions. Proficient in Kotlin, Java, Flutter, Kotlin Multiplatform, Clean Architecture, MVVM, SDK Development, REST APIs, and large-scale public sector and transit applications. Also experienced in MDM solutions (Headwind, MeshCentral), cryptography, MQTT and WebSocket communication, and AI/ML including deep learning, YOLO, and generative AI. Proven ability to deliver secure, scalable, and high-performance mobile solutions focused on reliability, maintainability, and user experience.

## Skills

- Android
- Flutter
- Android Jetpack
- Hilt / Dagger (DI)
- Retrofit
- Firebase
- TensorFlow Lite
- Facial Recognition
- Auto. Fare Collection (AFC)
- Machine Learning Integration
- Kotlin Coroutines
- Kotlin Multiplatform (KMP)
- POS Contactless Payments
- Deep Learning
- Image Classification
- Image Segmentation
- Generative AI
- Kotlin
- Dart
- Clean Architecture
- SDK Development
- WebSocket & Real-Time Comms
- Git
- OpenCV
- Anti-Spoofing Systems
- Payment Gateway Integration
- Google Maps
- RxJava
- MQTT
- MDM: Headwind, MeshCentral
- Android & Windows Agents
- Cryptography & Hashing
- YOLO
- Compose Multiplatform (CMP)
- Java
- Jetpack Compose
- MVVM
- REST APIs
- Socket.io
- CI/CD Pipelines
- Computer Vision
- NFC & Smart Card
- Offline-First Architecture
- Room
- Android Studio
- Python
- Pandas
- NLP & LLMs
- PCI DSS Compliance

## Experience | 6+ years

**MOBILE APPLICATION DEVELOPER / ANDROID DEVELOPER | PUNJAB INFORMATION TECHNOLOGY BOARD — Lahore** Dec 2021 – Present

Lead developer on enterprise-grade public sector applications including AFC systems, facial recognition attendance, waste management monitoring, and real-time transit platforms. Architected offline-first solutions, integrated ML models (TensorFlow Lite, OpenCV), built NFC & smart card features, and maintained CI/CD pipelines with Git and Firebase.

**ANDROID DEVELOPER | HORIZAM TECH — Lahore** Oct 2020 – Dec 2021

Developed and published multiple Android apps on the Play Store including NFC-based contact sharing (MetroTap), social networking (Find New Friends), and service marketplace apps. Worked with Firebase, Fragments, RecyclerViews, and Material Design.

**ANDROID DEVELOPER | TALKAGENT — Lahore** Feb 2020 – Oct 2020

Built a food delivery rider app (Twidlee Rider), a healthcare lab report app (Camden Care), and a Google Sheets API-driven utility. Gained experience with RESTful APIs, real-time order tracking, and Play Store deployment.

**ANDROID DEVELOPER INTERN | PUNJAB INFORMATION TECHNOLOGY BOARD — Lahore** Nov 2019 – Feb 2020

Developed a QR code generation/scanning app, a children's coloring app, and a celebrity image-guessing game independently. Gained foundational experience with Android Studio, Git, and app deployment.

## Projects

**Punjab Masstransit Authority | Punjab Information Technology Board** Apr 2022 – Present

https://play.google.com/store/apps/details?id=pk.pitb.punjab.masstransit.punjab_masstransit_authority

Real-time public transport information app for Punjab. Features live arrival/departure updates, AFC integration, network maps, disruption alerts, and journey planning. Built with MVVM, Clean Architecture, and offline-first support.

**Saaf Punjab — Facial Attendance | Punjab Information Technology Board** Jan 2023 – Mar 2023

https://play.google.com/store/apps/details?id=pk.pitb.gov.saafpunjabfacial

Enterprise facial recognition attendance system with five-pose worker registration, anti-spoofing, geo-fenced check-in, shift management, and 24-hour offline capability using TensorFlow Lite and OpenCV.

**GRMS | Punjab Information Technology Board** Jun 2023 – Jul 2023

https://play.google.com/store/apps/details?id=pk.pitb.gov.grms

Government Residences Management System — streamlines allotment, house reservation, resident profiling, and complaint resolution for government officers with a single-click experience.

**Saaf Punjab IRIS | Punjab Information Technology Board** Mar 2023 – May 2023

https://play.google.com/store/apps/details?id=pk.gov.pitb.saafpunjab.iris

Biometric (IRIS-based) monitoring platform for Waste Management Companies to evaluate daily attendance of supervisors and field workers.

**MetroTap | Horizam Tech** May 2021 – Dec 2021

https://play.google.com/store/apps/details?id=com.horizam.MetroTap

NFC-based instant contact and social media sharing app. Tap a smart card to share profiles, payment links, and websites. Published on Play Store.

**Camden Care | TalkAgent** Feb 2020 – Mar 2020

https://play.google.com/store/apps/details?id=com.camdencare.app

Mobile app for Camden Health System allowing users to securely access their lab reports. Published on Play Store.

**Auto Silent Application** Jul 2018 – Jul 2019

https://play.google.com/store/apps/details?id=com.asad.fareed

Final Year Project — auto-switches device ringer to silent mode based on user-defined locations and time schedules.

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## Education

**COMSATS Institute of Information Technology, Vehari** Bachelor of Sciences in Computer Science (2019) — GPA 3.6

**Government Post Graduate College, Burewala** FSC (2015)

**Quaid Public School, Burewala** Matriculation (2013)

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## Certifications

- Computer System Engineering (2013)
- PCI DSS Compliance Project — Certificate of Completion of Training, 360 Technologies (Pvt) Ltd. with PITB

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## Awards

- Best Cultural Representative (2016)